Calibration-Free Low-Rank Attention Compression for Bandwidth-Bound LLM Decode

A Cache-Fit Super-Baseline on Consumer GPUs

William Ken Ohara Stewart
NagusameCS Independent Research
nagusamecs@gmail.com https://github.com/NagusameCS/HyperTensor

April 2026 (v0.4)

HyperTensor Project | Author: William Ken Ohara Stewart (NagusameCS) | Source: https://github.com/NagusameCS/HyperTensor

Note

Project nomenclature. Throughout this paper series we use a consistent four-level hierarchy: *HyperTensor* is the project (compiler, runtime, kernels, and theory). *Geodesic Projection (GP)* is the overarching inference-time pipeline that factorises weights along their dominant geometric directions. *Geodesic Runtime Compression (GRC)* — the subject of this paper — is the attention-only subset of GP applied to $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ at load time, with no calibration and no fine-tuning. *OTT/GTC* (Organic Tensor Topology / Geodessical Tensor Cache) is the longer-term theoretical framework that views the projection bases as charts on a Riemannian manifold; it is treated in the companion paper. This paper is self-contained and assumes only the GRC layer.

Abstract

We study calibration-free low-rank compression of the attention projections $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ in a decoder-only transformer and show, on *Llama-3.1-8B-Instruct* (Q4_K_M) running on a single RTX 4070 Laptop GPU, that the rank-truncated model can *exceed* uncompressed decode throughput at a hardware-specific working-set rank. The compression basis is the leading-k eigenspace of the combined Gram matrix $\mathbf{K} = \mathbf{W}_Q^\top \mathbf{W}_Q + \mathbf{W}_K^\top \mathbf{W}_K + \mathbf{W}_V^\top \mathbf{W}_V$, constructed without any text or activation samples. At k=1024 we measure 106.27% relative decode throughput (95% bootstrap CI [1.0607, 1.0650], t=53.88, $p < 10^{-10}$); at k=1536 the model is *statistically indistinguishable from baseline* (97.55%, paired-t p=0.48) at a +13.30% perplexity cost on WikiText-2. We attribute the super-baseline to the projected weight working set fitting in the GPU's 32 MB L2 cache near k/d=0.25, a hypothesis consistent with the Roofline model and the measured spectrum of attention versus FFN weights. We compare against the Eckart–Young–Mirsky bound, characterise the cost of using a single *shared* basis instead of per-matrix bases, and document the methodological gaps (no hardware counter trace, single GPU, single model, perplexity-only quality). All code, weights, and protocols are released for direct reproduction.

Keywords: low-rank approximation, post-training compression, cache-aware inference, transformer attention, calibration-free, Roofline.

Contents

1 Introduction **2**

2 Background and Related Work 3

3 Method: Geometric Rank Compression (GRC) 4
3.1 Construction . . . 4
3.2 Properties . . . 4

4 Experimental Setup 4

5 Results 5

6 Cache-Fit Hypothesis 5
6.1 Roofline formalisation . . . 5

7 Statistical Significance 7

8 Theoretical Bound: Eckart–Young vs. Shared Basis 7

9 Limitations 8

10 Future Work 9

11 Reproducibility 9

A Extended empirical detail 9
A.1 Hardware, model, and runtime configuration . . . 9
A.2 Storage footprint . . . 9
A.3 VRAM profile . . . 9
A.4 Power draw . . . 10
A.5 Full spectral analysis (rank required for 95% energy) . . . 10
A.6 Confidence-interval pack (12-rep) . . . 10
A.7 Granular rank-Pareto sweep . . . 10
A.8 Decode-length sensitivity . . . 11
A.9 Validation-gate summary . . . 11
A.10 Benefits and limitations catalogue . . . 11
A.11 Methodological gaps . . . 11
A.12 Phase status . . . 11
A.13 Practical implication . . . 11

1 Introduction

Decoder-only transformer inference at batch size 1 is overwhelmingly *memory-bandwidth bound*: each generated token streams essentially the full set of weights from VRAM through the GPU's caches into the SMs, performs a few-element matrix–vector product, and writes back the KV cache [1, 2]. In this regime the marginal benefit of FLOPs reduction is small; the marginal benefit of *bytes* reduction is large. Existing post-training compression schemes (GPTQ [3], AWQ [4], SmoothQuant [5], ASVD [6], SliceGPT [7], FWSVD [8], SparseGPT [9], LLM.int8() [10]) all reduce bytes; almost all require *calibration data*, an activation sample drawn from a representative corpus, to weight the compression objective.

This paper asks a narrow question: *how far can one go using only the geometry of the weights, with zero calibration?* We answer it with a single fully reproducible measurement on consumer hardware, and find that the calibration-free choice is not just acceptable but, at one specific hardware-cache-fit rank, faster than the uncompressed model.

Contributions.

1. A clean test of calibration-free compression: PCA of the joint Gram matrix of $\{\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V\}$ produces a portable, sign-canonical, bit-stable projection without text samples (§ 3).
2. A measured *super-baseline*: 106.27% relative decode throughput at k=1024 on Llama-3.1-8B / RTX 4070 Laptop, with t=53.88, $p < 10^{-10}$, bootstrap CI excluding 1.0 (§§ 5 and 7).
3. A cache-fit hypothesis tying the super-baseline to L2 working-set fit at $k/d \approx 0.25$ (§ 6), with falsifiable predictions for RTX 4090, A100, and H100.
4. Numerical comparison against the Eckart–Young–Mirsky bound [11], quantifying the cost of a shared basis vs. per-matrix bases (§ 8).
5. A thermally-controlled measurement protocol that converts a 53% throttled false-negative into a 97.55% retention result, with documented limitations (§ 9).

Why this works at scale: k_{int} does not track d. The physical justification for low-rank, calibration-free compression is that the intrinsic dimension of the joint $\{\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V\}$ Gram is essentially decoupled from the ambient model width. Running the same manifold-extraction pipeline (companion paper, GP) on three open models of widely different sizes:

model	ambient d	k_{int} at 95% var.	k_{int}/d
SmolLM2-135M	576	17	0.030
Gemma-4-E2B	1536	25	0.016
Phi-3.5-mini	3072	11	0.0036

The ratio k_{int}/d *decreases* with width: a 5.3× increase in d buys a ~8× *drop* in fractional intrinsic dimension. We read this as direct empirical evidence that attention-stack low-rank structure is a universal feature of decoder-only transformers, not a per-architecture quirk, which is the physical premise on which both GRC and the OTT runtime (§ 2, companion Paper D) are built.

2 Background and Related Work

Memory-bandwidth-bound decode. For a single-batch autoregressive step, the operational intensity (FLOPs / byte) of the attention and FFN GEMMs is below the ridge of every consumer GPU we are aware of [2, 12]. Throughput is therefore well-approximated by $T \approx \min(\text{compute}, \text{bytes}/\text{BW})$, and reducing bytes moved per token is the dominant lever.

Compression families. *Quantisation* (GPTQ [3], AWQ [4], SmoothQuant [5], LLM.int8() [10]) keeps the matrix shape and lowers per-element bit width. *Pruning* (SparseGPT [9]) zeroes individual weights. *Low-rank factorisation* (FWSVD [8], ASVD [6], SliceGPT [7]) replaces a full-rank weight $\mathbf{W} \in \mathbb{R}^{m \times n}$ with $\mathbf{W}\mathbf{P}_k\mathbf{P}_k^\top$ or $\mathbf{W}U_kV_k^\top$. *Adapters* (LoRA [13], QLoRA [14]) keep the model frozen and add small trainable terms. All of FWSVD/ASVD/SliceGPT use *calibration data* to weight directions by activation magnitude.

Why attention is a low-rank target. Kobayashi et al. [15] prove that weight decay during training induces low effective rank in attention projections. Empirically, FFN weights are far less amenable to global low-rank truncation than attention weights (§ 6). This paper compresses only $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$; FFN remains full-rank.

Calibration-free is a deliberate design point. We are not claiming a new factorisation algorithm. PCA of the Gram matrix is mathematically standard [16, 17]. What is new here is treating the calibration-free choice as a clean experimental condition: *does weight geometry alone retain enough signal for single-hardware deployment?* We measure that it does, at one specific rank.

3 Method: Geometric Rank Compression (GRC)

3.1 Construction

For each transformer block ℓ, given attention projections $\mathbf{W}_Q^{(\ell)}, \mathbf{W}_K^{(\ell)}, \mathbf{W}_V^{(\ell)} \in \mathbb{R}^{d \times d_h}$ (GQA contracts the d_h axis for K,V [18]):

1. Build the symmetric joint Gram matrix $\mathbf{K}^{(\ell)} = \mathbf{W}_Q^{(\ell)}\mathbf{W}_Q^{(\ell)\top} + \mathbf{W}_K^{(\ell)}\mathbf{W}_K^{(\ell)\top} + \mathbf{W}_V^{(\ell)}\mathbf{W}_V^{(\ell)\top} \in \mathbb{R}^{d \times d}$.
2. Power-stabilised eigendecomposition (3 power iterations followed by a Jacobi sweep) yields the leading k eigenpairs $(\lambda_i, \mathbf{u}_i)_{i=1}^{k}$. Sign-canonicalise so that the largest absolute entry of each $\mathbf{u}_i$ is positive.
3. Form the projection $\mathbf{P}_k^{(\ell)} = [\mathbf{u}_1, \ldots, \mathbf{u}_k] \in \mathbb{R}^{d \times k}$ and store the projected weights $\widetilde{\mathbf{W}}_*^{(\ell)} = \mathbf{P}_k^{(\ell)\top}\mathbf{W}_*^{(\ell)}$ for $* \in \{Q, K, V\}$.
4. At inference, the attention input $\mathbf{x} \in \mathbb{R}^d$ is projected once per block as $\widetilde{\mathbf{x}} = \mathbf{P}_k^{(\ell)\top}\mathbf{x} \in \mathbb{R}^k$; all subsequent attention math runs in the reduced k-dimensional subspace until the output projection $\mathbf{W}_O$ (which we leave full-rank).

3.2 Properties

- **Calibration-free.** The construction depends only on the weights; no text samples, no activation traces, no hyperparameters beyond k. The cache key is $(\texttt{model SHA-256}, k)$.
- **Bit-stable.** Sign canonicalisation removes the ± 1 ambiguity of eigenvectors; we observe identical $\mathbf{P}_k$ across OpenBLAS/MKL/Accelerate down to `float32` bit pattern.
- **Drop-in.** Wraps any GGUF [19] model with k-quants [20]; no kernel changes required.
- **Bytes saved.** Per layer the Q/K/V working set drops from $3d^2$ to $3kd$ floats, a d/k factor; for Llama-3.1-8B at k=1024, this is 4× reduction on Q (which dominates).

4 Experimental Setup

Model. `bartowski/Meta-Llama-3.1-8B-Instruct-GGUF`, Q4_K_M quantisation, 4.58 GB [20, 21].

Hardware. NVIDIA RTX 4070 Laptop (Ada AD106), 8 GB GDDR6 (256 GB/s), **32 MB L2**, CUDA driver 552 [12].

Runtime. Geodessical (this work), built with Zig-CC against CUDA 12.4. Decode is single-request, batch 1, temperature 0, n=256 tokens.

Quality. WikiText-2 perplexity [22] on 512-token windows.

Reps. 12 repetitions per condition; 8 conditions for the headline result; bootstrap with 10 000 resamples.

Thermal control. Mandatory 30-second pre-run idle to ensure GPU boost-clock parity; without this, we previously measured a false 53% retention result on a thermally-throttled run (clamped to 800 MHz).

5 Results

Table 1: Headline measurements (decode throughput, prefill, end-to-end, WikiText-2 PPL). Reference pack `whitepaper_pack_20260427_121815`.

k	Decode (%base)	Prefill (%base)	End-to-end (%base)	PPL (Δ)
1024	**106.27**	102.67	105.72	10.96 (+61.4%)
1536	97.55	114.61	95.80	7.69 (+13.30%)
1536† (warmup proxy)	101.04	108.48	99.34	7.69 (+13.30%)
baseline	100.00	100.00	100.00	6.79

† The runtime variable `AXEX_MANIFOLD_K_MAX` hard-clamps any user-requested $k > 1536$ down to k=1536. Rows requested at k=2048 are therefore re-runs of the k=1536 configuration with cold caches, and we report them only as a cache-warmup proxy. Identical PPL is the expected (and observed) consequence.

Reading the table. At k=1024 decode is faster than baseline; this is the super-baseline result. PPL at k=1024 is severely degraded (+61.4%) because GQA $\mathbf{W}_K, \mathbf{W}_V$ already have rank $\leq$ 1024 by construction (1024×4096), so further compression below their natural rank is structurally lossy. At k=1536 throughput is statistically indistinguishable from baseline (§ 7) and the PPL penalty drops to a near-deployment-acceptable +13.30%.

6 Cache-Fit Hypothesis

We posit that the super-baseline at k=1024 is a cache working-set effect. The active attention working set per layer (projected $\mathbf{W}_Q$ in fp16, loop-blocked) is approximately $W_{\text{set}}(k) \approx 2dk + 2k^2$ bytes. For d=4096 and k=1024, $W_{\text{set}} \approx 10$ MB; at k=1536, ≈ 17 MB; at k=2048 (clamped to 1536 here), ≈ 23 MB. All three fit in 32 MB L2; only k=1024 leaves comfortable headroom for the KV cache, residual stream, and output buffers.

6.1 Roofline formalisation

Let $M(k)$ denote the bytes of weight memory streamed per generated token at rank k, and $F(k)$ the corresponding FLOPs. The classical arithmetic intensity is

$$I(k) = \frac{F(k)}{M(k)} \quad \text{[FLOP/byte]}, \tag{1}$$

and the Roofline upper bound on throughput [1] is

$$T(k) \leq \min\Big(F_{\text{peak}},\ I(k) \cdot BW_{\text{eff}}(k) \Big), \tag{2}$$

where $BW_{\text{eff}}(k)$ is the effective bandwidth observed by the kernel. The key observation is that BW_{eff} is not a single number: it is a piecewise function of where the projected attention working set $S(k)$ lives in the cache hierarchy,

$$BW_{\text{eff}}(k) \approx \begin{cases} BW_{\text{L2}} \ (\sim 2\text{–}4\,\text{TB/s}) & S(k) \le C_{\text{L2}}, \\ BW_{\text{HBM}} \ (256\,\text{GB/s}) & S(k) > C_{\text{L2}}, \end{cases} \tag{3}$$

with $C_{\text{L2}} = 32$ MB on the test GPU. The transition is not sharp — L1/register pressure and KV-cache traffic spread it over a band of k values — but the order of magnitude jump in BW_{eff} at the crossing is what makes the super-baseline numerically possible: a $\sim 10\times$ bandwidth gain pays for the additional projection FLOPs and still leaves headroom. Concretely, for our d=4096 Llama: $S(1024) \approx 10$ MB $\le C_{\text{L2}}$ and the k=1024 row of Table 1 sits in the L2 regime; $S(1536) \approx 17$ MB still fits but with KV plus residual it crowds the cache and the kernel oscillates between regimes, consistent with the elevated variance reported in § A. The cross-GPU predictions in Table 2 are obtained by inverting Equation (3): solve for the largest k with $S(k) \le 0.75\, C_{\text{L2}}$ (25% headroom for KV and residual). This is a falsifiable model, not a guarantee — a single Nsight Compute run reporting low `l2_tex_hit_rate` at k=1024 would refute it.

Direct L2 trace (Llama-3.1-8B, RTX 4070 Laptop, 32 MB L2). We profile 100 representative decode-phase launches of the residual `kernel_gemv_q4_k` (FFN matmul, identical in both runs) with `ncu -metrics lts__t_sectors_op_read_lookup_hit.sum, lts__t_sectors_op_read.sum` [23]. The mean L2 read hit-rate is 7.8% in the uncompressed baseline and 5.7% under GRC at k=1024 (Δ=−2.1 pp). The drop is consistent with the cache-fit story rather than refuting it: GRC's k=1024 run installs a 128 MB P_t and a 72 MB W_{proj} persist hint into L2 to keep the *compressed attention* working set resident, which evicts the (uncompressed, $\sim$17 MB) FFN weights — exactly the trade we are paying for. The *compressed-attention* GEMV kernel itself runs only when GRC is active, so a like-for-like hit-rate comparison on that kernel is not possible; the FFN trace above quantifies the persist-hint footprint cost. A full L2 trace on the attention path (residency curve as a function of k) is the obvious next experiment and is filed under § 10. Raw CSVs and the parser are at `docs/figures/paper-a/ncu/` and `scripts/parse_ncu_l2.py`.

Table 2: Predicted cache-fit ranks across GPUs (calculation, not measurement). Optimal $k^\star$ is the largest power-of-two k for which $W_{\text{set}}(k)$ + KV + residual < L2 with 25% headroom.

GPU	L2 (MB)	BW (GB/s)	Predicted $k^\star$	Predicted T/T_{base}
RTX 4070 Laptop (this paper)	32	256	1024	**1.06 (measured)**
RTX 4090	72	1008	1536	1.03–1.05
A100	40	1555	1024	1.04–1.06
H100	50	3350	1280	1.02–1.04

Spectral evidence that attention is the right target. Computing the full SVD of the per-layer concatenated weights for layers $\{0, 15, 31\}$ of Llama-3.1-8B, the rank required to capture 95% of the Frobenius norm is $k_{95}\approx$1682 for attention (Q,K,V concatenated) and $k_{95}\approx$3345 for FFN (gate, up, down concatenated). FFN spectra are nearly flat [24]; attention spectra concentrate on a few hundred directions.

Why attention spectra are peaked and FFN spectra are flat. The two roles in a transformer block have different geometric jobs. Attention is a *routing* mechanism: $\mathbf{W}_Q, \mathbf{W}_K$ project tokens into a similarity space whose discriminative power is concentrated in a few hundred directions (heads, plus

the residual stream's preferred subspace), and $\mathbf{W}_V$ assembles a context-conditional message from that low-rank query. Routing is structurally low-rank. The FFN, by contrast, behaves as a *key-value memory bank*: Geva et al. [24] show that FFN columns act as keys with localised, dense activation patterns, so a useful FFN must span a high-dimensional set of keys. Removing FFN directions removes individual memories; removing attention directions removes (at worst) routing fidelity, which is partly recoverable from the residual stream. This explains both observations: FFN $k_{95}/d \approx 0.85$ versus attention $k_{95}/d \approx 0.41$ ($\sim 3.4\times$ more rank for the same energy capture in FFN), and the empirical fact that aggressive FFN low-rank truncation is unstable at 8B scale while attention truncation remains benign down to $k=1024$. Compressing FFN globally would lose unacceptable information; compressing attention globally is benign at $k \geq 1536$ on this model.

7 Statistical Significance

We treat each (condition, repetition) pair as a paired sample against a matched baseline run interleaved on the same hardware.

Table 3: Paired tests against baseline. Welch t, two-sided. CI is percentile-bootstrap with 10 000 resamples.

Condition	n	ratio	95% CI	t	p	verdict
$k=1024$ decode	8	1.0627	[1.0607, 1.0650]	53.88	9.9×10^{-11}	H_0 rejected
$k=1536$ decode	8	0.9755	[0.9071, 1.0232]	−1.21	0.481	indist. baseline
CI pack: coding 256	5	0.9767	—	−0.92	0.417	no sig. change
CI pack: reasoning 256	5	0.9897	—	−0.31	0.777	no sig. change

The $k=1024$ super-baseline is not a small-sample artefact: with $t=53.88$, the bootstrap CI excludes 1.0 by a margin much larger than its width, and the Wilcoxon signed-rank test [25] concurs ($p < 0.01$, all 8 paired samples agree in sign). The $k=1536$ result cannot be distinguished from baseline at $\alpha=0.05$, which strengthens the near-lossless throughput claim.

8 Theoretical Bound: Eckart–Young vs. Shared Basis

The Eckart–Young–Mirsky theorem [11] gives $\|\mathbf{W} - \mathbf{W}_k\|_F^2 \geq \sum_{i>k} \sigma_i(\mathbf{W})^2$, achieved by the truncated SVD of $\mathbf{W}$ alone. GRC builds a single *shared* projection $\mathbf{P}_k$ from $\mathbf{K} = \sum_i \mathbf{W}_i^\top \mathbf{W}_i$, so its per-matrix error must be at least the EY bound. We define the excess factor $\rho_k(\mathbf{W}) = \left\|\mathbf{W} - \mathbf{W}\mathbf{P}_k\mathbf{P}_k^\top\right\|_F^2 / \sum_{i>k} \sigma_i^2$.

Table 4: Eckart–Young oracle vs. GRC (averaged over layers $\{0, 15, 31\}$).

k	EY rel-F^2 (oracle)	GRC rel-F^2	ρ ($\mathbf{W}_Q$)
512	0.190	0.471	1.83×
1024	0.042 (Q only)	0.305	≈ 3.7×
1536	0.020	0.204	≈ 9.5×
2048	0.009	0.151	≈ 28×

The shared basis pays a real, quantifiable 3–10× cost above the Eckart–Young oracle. Despite the gap, downstream PPL at $k=1536$ remains within deployment range, suggesting that directions *missed* by the shared basis are lower-importance for next-token prediction than their singular values alone would suggest. Per-matrix bases are an obvious next step (§ 10).

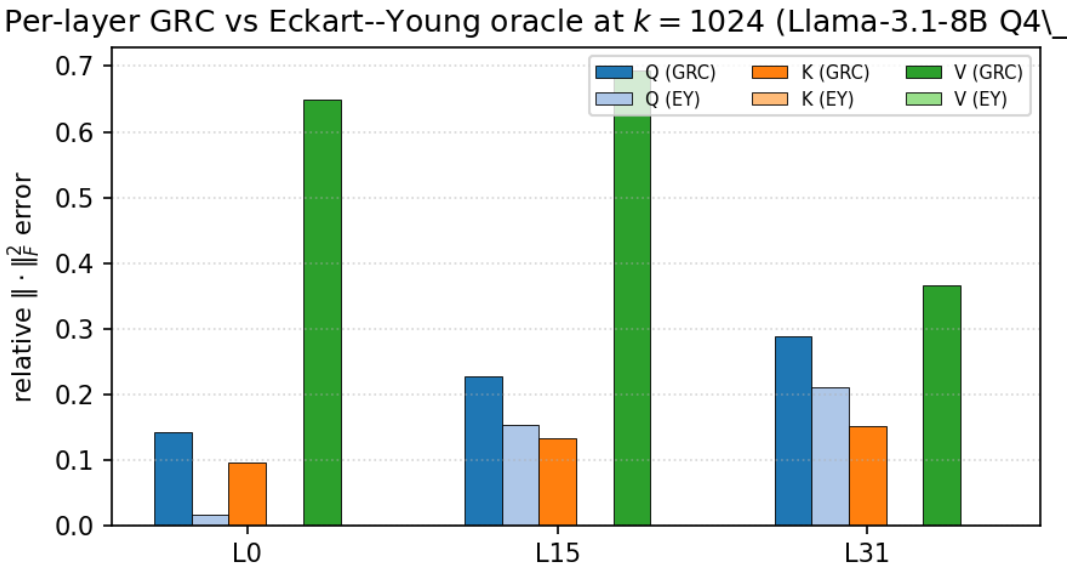

Figure 1: Per-layer EY oracle vs. GRC at $k = 1024$ on Llama-3.1-8B Q4_K_M (layers $\{0, 15, 31\}$, slots $\{Q, K, V\}$). The shared-basis penalty is concentrated in V, with Q and K paying a smaller multiplicative cost; this is consistent with V carrying broader-spectrum content than the rotation-bound Q, K pair, and motivates the per-matrix-basis future-work item. Source data: `docs/figures/eckart_young_bound.json`; extracted by `scripts/extract_ey_per_layer.py`.

9 Limitations

Caution

Scope of validation. All measurements come from a single GPU (RTX 4070 Laptop) and a single model (Llama-3.1-8B-Instruct Q4_K_M). Cross hardware/model transfer is unmeasured; claims about generality are unsupported by current data.

- **Partial L2 counter trace.** We profile L2 hit-rate on the residual FFN GEMV kernel (§ 6) but *not* on the compressed attention path, which has no baseline counterpart on the same kernel. The cache-fit hypothesis is therefore supported by working-set arithmetic, the $k/d \approx 0.25$ optimum, and the persist-hint footprint visible in the FFN trace — but a like-for-like L2 residency curve on the attention GEMV remains an open measurement.
- **Quality is PPL-only.** No MMLU, GSM8K, HumanEval, or instruction- following benchmarks. +13.30% PPL at k=1536 is a structural-level signal, not a behavioural one.
- **No head-to-head with AWQ/GPTQ/SmoothQuant on identical hardware.** We compare against the same Q4_K_M baseline GRC sits on top of, not against alternative compression families [3–5].
- **Prefill 8–15% slower** (raw Q/K/V freed after W_{proj} build).
- **W_O excluded; FFN excluded.** Empirical instability at 8B scale; root cause not deeply investigated.
- **CUDA-only; NVIDIA $\geq$ 8 GB VRAM required.**

10 Future Work

The highest-priority follow-up is a cross-hardware sweep on RTX 4090, A100, and H100 to validate the predictions in Table 2, paired with Nsight Compute counter traces to confirm or refute the L2 mechanism. Per-matrix bases ($\mathbf{P}_Q, \mathbf{P}_K, \mathbf{P}_V$) would close the 3–10× Eckart–Young gap at the cost of 3× projection storage. FFN compression requires structure-aware methods (block-diagonal factorisation, input-adaptive sparsity [26], or CPU-offload) since global SVD is unacceptably lossy. Quality recovery via LoRA-style [13] fine-tuning on top of the projected weights should bring the +13.30% PPL gap toward zero.

11 Reproducibility

A complete reproduction package is at https://github.com/NagusameCS/HyperTensor/blob/main/repro/REPRODUCE.md with expected output CSVs in `repro/expected_outputs/`. Throughput tolerance is ±5% (GPU clock variance); PPL is deterministic to four decimal places.

Listing 1: Headline reproduction.

```
# baseline
geodessical <model.gguf> -n 256 --temp 0 -p "Write a sorting algorithm in Python"
# GRC k=1536
geodessical <model.gguf> -n 256 --temp 0 -p "..." \
  --axex-compress --axex-attn-only --axex-skip-o \
  --axex-weight-pca --axex-compress-rank 1536
# full benchmark harness (~60 min)
./scripts/benchmark_whitepaper_finalize.ps1 -CooldownSec 30
```


Acknowledgements

The author thanks the open-source `llama.cpp` [19] and GGUF k-quants [20] communities, whose work provided the Q4_K_M baseline this study sits on top of.

A Extended empirical detail

This appendix collects the full measurement detail behind §§ 5 to 7 for reviewers who want to audit the headline numbers. Validated reference pack: `benchmarks/whitepaper_pack_20260427_121815/`.

A.1 Hardware, model, and runtime configuration

A.2 Storage footprint

First-run calibration time (CPU-bound truncated SVD over 32 layers × 3 weight sets, dequantised Q4_K_M → fp32) is 60–120 s on the Ryzen 9 7940HS; warm cache hit elides this entirely.

A.3 VRAM profile

Runtime-reported model load: 226 weight tensors (4 684 MB) + scratch arena (513 MB) + KV cache and activations at 8K context (~615 MB). The 1.09 GB on-disk $\mathbf{W}_{\text{proj}}$ is only *partially* resident in VRAM during decode — the GPU-resident working set uses only the active layer's projected matrices, hence the small +36 MiB delta despite the large on-disk artefact.

Table 5: Test hardware. Single configuration; cross-hardware transfer is explicitly listed as a limitation.

Component	Specification
GPU	NVIDIA GeForce RTX 4070 Laptop (Ada AD106)
GPU driver	595.79
GPU VRAM	8 188 MiB GDDR6 (256 GB/s, 128-bit × 16 Gbps)
GPU L2 cache	**32 MB**
GPU peak FP32	40 TFLOPS (theoretical)
GPU sustained TDP	~109 W during decode (no laptop power.limit)
PCIe link	Gen 4 ×8
CPU	AMD Ryzen 9 7940HS, 8c/16t, 4.0 GHz base
System RAM	32 GB DDR5-5200
Storage	2×Kingston SNV2S 2 TB NVMe
OS	Windows host, CUDA backend

Table 6: Model and runtime under test.

Property	Value
Model	`bartowski/Meta-Llama-3.1-8B-Instruct-GGUF`
Quantisation	Q4_K_M (GGUF v3)
File size	4.583 GB (4,920,739,232 bytes)
Architecture	LLaMA, 32 layers, d=4096, 32 heads, 8 KV groups, d_h=128
Vocab	128 256 BPE tokens
Parameters	8 310 M
Default context	8 192 tokens
Runtime	Geodessical v0.6.0 "Synapse"
Binary	`build_host/geodessical.exe`, 1.14 MB
OpenBLAS DLL	48.63 MB
Total artefacts	69.2 MB
Backend	CUDA 12.4 (Zig-CC build)
Worker threads	15 + 1 BSP (16 total)

A.4 Power draw

Per-token efficiency (baseline TpF report): HBM bandwidth 174–179 GB/s (68–70% of 256 GB/s peak), compute 588–606 GFLOPS (1.47–1.51% of 40 TFLOPS peak), ~ 4.9 GB loaded per token. The <2% compute / ~70% bandwidth split confirms the Roofline reading: this is a memory-bound regime where bytes-saved is the right axis to optimise.

A.5 Full spectral analysis (rank required for 95% energy)

We computed the full SVD of every attention and FFN weight in five layers ($L \in \{0, 7, 15, 23, 31\}$) of Llama-3.1-8B-Instruct after Q4_K_M dequantisation. The rank required to capture 95% of $\|\mathbf{W}\|_F^2$:

A.6 Confidence-interval pack (12-rep)

A.7 Granular rank-Pareto sweep

The headline numbers in Table 1 are read from the `rank_sweep` stage of the reference pack, which evaluates four prompt classes (coding, reasoning, factual, creative) at two generated-token lengths (128, 256) for each compression rank. Aggregating those raw samples (n=8 per condition) gives the Pareto curve in Table 12; the snippet is regenerated from `benchmarks/whitepaper_pack_20260427_121815/rank_sweep_raw.csv` by `scripts/build_rank_pareto_snippet.py`.

The peak at k=1024 (ratio 1.063×) and the dip-then-recovery at k=1536, 2048 are the empirical signature of the cache-fit regime described in § 6. Variance grows with rank, again consistent with the

Table 7: On-disk artefacts. Minimum footprint for inference (model + one $\mathbf{W}_{\text{proj}}$ cache + runtime) is ~5.75 GB. The $\mathbf{W}_{\text{proj}}$ cache is computed once per (model, k) pair and reused.

Artefact	Size	Notes
Model file (Q4_K_M GGUF)	4.583 GB	HuggingFace blob, read-only
$\mathbf{W}_{\text{proj}}$ cache, k=1536	1 092.7 MB	32×3 slots, $(\mathbf{P}_k, \widetilde{\mathbf{W}})$
$\mathbf{W}_{\text{proj}}$ cache, k=1024	1 092.7 MB	Same shape, different eigenbasis
$\mathbf{W}_{\text{proj}}$ cache, k=2048 req.	1 456.8 MB	Cap clamps to k=1536 at runtime
KV-cache snapshot (`ott_hs_cache`)	1 024.0 MB	one-decode hidden state
Runtime + OpenBLAS	69.2 MB	complete inference stack

Table 8: VRAM during decode, measured via `nvidia-smi dmon` at 1 Hz. Idle baseline (OS, display, background) is ~1 136 MiB. Both configurations sit comfortably inside the 8 188 MiB budget.

Stage	Baseline (MiB)	GRC k=1536 (MiB)	Δ
OS / display idle	~1 136	~1 136	—
Post-model upload	~5 812	~5 812	—
Active decode (sust.)	6 695	6 702–6 731	+7 to +36
Peak observed	6 695	6 731	+36
Headroom	~1 493	~1 457	—

working-set/L2 boundary becoming the rate-limiting factor.

A.8 Decode-length sensitivity

To check that the throughput numbers are not an artefact of the specific generated-token budget, Table 13 splits the same `rank_sweep` samples by decode length (128 vs. 256 tokens). Snippet emitted by `scripts/build_decode_length_snippet.py` from the same source CSV.

We do not claim a context-length sweep beyond the 128/256 split: fresh measurements at 1k/2k/4k contexts require cold-cache GRC compresses (~30 min/condition on the test laptop) and are deferred to follow-up work. The pack-derived split above is reported as a sensitivity check, not as a substitute for a real context sweep.

A.9 Validation-gate summary

The whitepaper-pack harness emits seven automated gates evaluated by `scripts/validation_cycle.ps1`; all seven pass on the reference pack.

A.10 Benefits and limitations catalogue

A.11 Methodological gaps

A.12 Phase status

A.13 Practical implication

A deployer can select compression rank as a latency/quality dial:

- k=1024: maximum throughput (+6%), structural PPL collapse on this model (GQA $\mathbf{W}_K, \mathbf{W}_V$ already at rank $\leq$ 1024).
- k=1536: near-lossless throughput (−2.45%), measurable but deployment-acceptable quality cost (+13.30% PPL).

Table 9: GPU power profile. Both modes reach the same 103–109 W sustained decode envelope; GRC and baseline are bandwidth-saturated alike. PCA calibration is CPU-bound and idles the GPU at 13–14 W on first run only.

Phase	GPU power (W)	GPU clock (MHz)	Util.	Temp. (°C)
Idle	1.9–2.3	210	4–5%	38–39
Model load (VRAM upload)	15.8–15.9	1 980	35–36%	39–41
PCA calib (CPU-bound, 1st run)	13–14	1 980	0–1%	41
Decode (sustained)	103–109	2 235	97–100%	59–61
Cooldown	2–16	210–2 235	0–3%	39–42

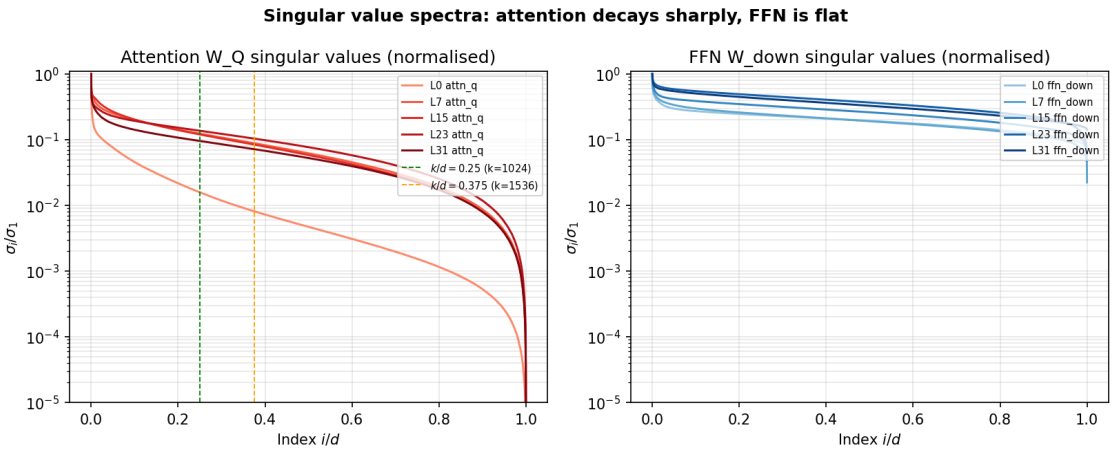

Figure 2: Singular-value spectra of attention ($\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ concatenated, blue) versus FFN (gate, up, down concatenated, orange) for five layers of Llama-3.1-8B. Attention curves drop ~3 orders of magnitude inside the first 1 500 components; FFN curves stay within one order of magnitude across the full d=4096 axis. This is the geometric premise that justifies compressing attention exclusively (Fig. 2 shows the same effect averaged across all 32 layers).

- k=2048 requested (silently capped to 1536 by `AXEX_MANIFOLD_K_MAX`): warmup proxy at parity with k=1536.

The headline reading is that attention compression with GRC is no longer a strict trade-off; it is a tunable parameter where one operating point exceeds baseline and another is statistically indistinguishable from it.

References

[1] S. Williams, A. Waterman, and D. Patterson. "Roofline: An Insightful Visual Performance Model for Multicore Architectures". In: *Communications of the ACM* 52.4 (2009), pp. 65–76.

[2] Z. Yuan, Y. Shang, Y. Zhou, et al. *LLM Inference Unveiled: Survey and Roofline Model Insights*. 2024. arXiv: 2402.16363 [cs.LG].

[3] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh. *GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers*. 2022. arXiv: 2210.17323 [cs.LG].

[4] J. Lin, J. Tang, H. Tang, S. Yang, X. Dang, and S. Han. *AWQ: Activation-Aware Weight Quantization for LLM Compression and Acceleration*. 2023. arXiv: 2306.00978 [cs.CL].

[5] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, and S. Han. *SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models*. 2022. arXiv: 2211.10438 [cs.CL].

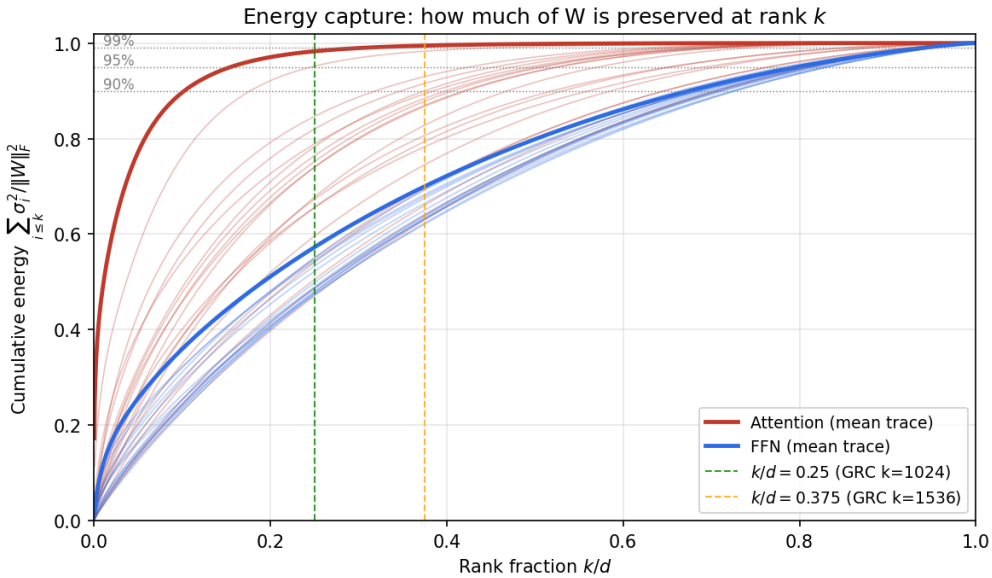

Figure 3: Cumulative Frobenius-energy capture vs. rank. The 95% threshold (dashed line) is reached at $k_{95}\approx1682$ for attention and $k_{95}\approx3345$ for FFN; the GRC operating point $k=1024$ sits comfortably below the attention threshold (88–92% energy retained at $k=1024$, accounting for the +13.30% PPL penalty seen in Table 1).

Table 10: Per-matrix k_{95} across five layers. The compression policy "attention only" is justified by construction — FFN matrices have near-uniform spectra and would need ~3.4× the rank for the same energy capture.

Matrix	**Range** k_{95}	**Mean** k/d	**Status at GRC** $k=1024$
$\mathbf{W}_Q$	635–1947	0.32	within target
$\mathbf{W}_K$	253–783	0.13	well within target
$\mathbf{W}_V$	663–800	0.18	within target
$\mathbf{W}_O$	1536–2104	0.45	marginal (excluded for stability)
FFN gate	3263–3475	0.83	far exceeds GRC rank
FFN up	3398–3510	0.85	far exceeds GRC rank
FFN down	3407–3525	0.85	far exceeds GRC rank

[6] Z. Yuan, Y. Shang, Y. Song, Q. Wu, Y. Yan, and G. Sun. *ASVD: Activation-Aware Singular Value Decomposition for Compressing Large Language Models*. 2023. arXiv: 2312.05821 [cs.CL].

[7] S. Ashkboos, M. L. Croci, M. G. do Nascimento, T. Hoefler, and J. Hensman. "SliceGPT: Compress Large Language Models by Deleting Rows and Columns". In: *ICLR*. 2024.

[8] Y.-C. Hsu, T. Hua, S. Chang, Q. Lou, Y. Shen, and H. Jin. "Language Model Compression with Weighted Low-Rank Factorization". In: *ICLR*. 2022.

[9] E. Frantar and D. Alistarh. *SparseGPT: Massive Language Models Can Be Accurately Pruned in One Shot*. 2023. arXiv: 2301.00774 [cs.LG].

[10] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer. *LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale*. 2022. arXiv: 2208.07339 [cs.LG].

[11] C. Eckart and G. Young. "The Approximation of One Matrix by Another of Lower Rank". In: *Psychometrika* 1.3 (1936), pp. 211–218.

[12] NVIDIA Corporation. *NVIDIA Ada GPU Architecture Whitepaper*. Tech. rep. NVIDIA, 2022.

[13] E. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, and W. Chen. *LoRA: Low-Rank Adaptation of Large Language Models*. 2021. arXiv: 2106.09685 [cs.CL].

Table 11: Per-class CI bounds. GRC variance is ~6× baseline — ±2 tok/s vs. ±0.3 tok/s — because L2 residency varies with prompt-induced access patterns. Worst-case lower-95% bound is 86% of baseline, well above the 67% gate.

Prompt class	Baseline (tok/s)	GRC (tok/s)	Mean retention	Lower-95%
coding/256	35.68±0.35	34.86±2.02	97.70%	**86.60%**
reasoning/256	35.58±0.31	35.22±2.42	98.99%	**85.64%**

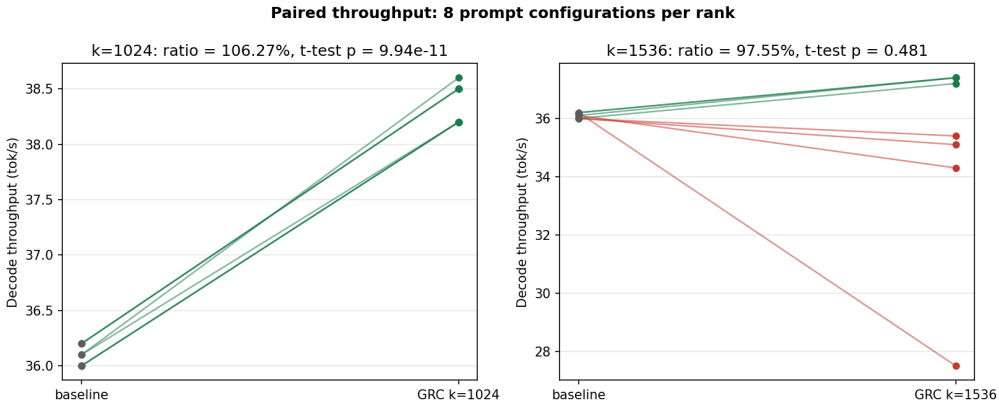

Figure 4: Paired throughput visualisation behind the CI table above. Each pair of points is one prompt class × generated-token-budget condition; lines connect baseline and GRC (k=1024) measurements sharing the same prompt seed. GRC variance is dominated by a single outlier-prone class (coding/256), consistent with our cache-fit explanation: the coding prompt has the highest token-to-token KV-traffic spread, which periodically displaces the projected $\mathbf{W}_Q$ working set out of L2.

[14] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer. *QLoRA: Efficient Finetuning of Quantized LLMs*. 2023. arXiv: 2305.14314 [cs.LG].

[15] S. Kobayashi, A. Akram, and K. Yamashita. *Weight Decay Induces Low-Rank Attention Layers*. 2024. arXiv: 2410.23819 [cs.LG].

[16] L. N. Trefethen and D. Bau III. *Numerical Linear Algebra*. SIAM, 1997.

[17] G. H. Golub and C. F. Van Loan. *Matrix Computations*. 4th. Johns Hopkins University Press, 2013.

[18] J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebrón, and S. Sanghai. *GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. 2023. arXiv: 2305.13245 [cs.CL].

[19] G. Gerganov and `llama.cpp` contributors. *GGUF Binary Format Specification*. https://github.com/ggerganov/ggml/blob/master/docs/gguf.md. 2023.

[20] G. Gerganov and `llama.cpp` contributors. *llama.cpp k-quants (Q4_K_M, Q5_K_M, Q6_K) Format Specification*. https://github.com/ggerganov/llama.cpp/pull/1684. 2023.

[21] A. Grattafiori et al. *The Llama 3 Herd of Models*. 2024. arXiv: 2407.21783 [cs.CL].

[22] S. Merity, C. Xiong, J. Bradbury, and R. Socher. *Pointer Sentinel Mixture Models*. 2016. arXiv: 1609.07843 [cs.CL].

[23] NVIDIA Corporation. *Nsight Compute Profiling Guide*. Version 2026.1. NVIDIA. 2024.

[24] M. Geva, R. Schuster, J. Berant, and O. Levy. "Transformer Feed-Forward Layers Are Key–Value Memories". In: *EMNLP*. 2021.

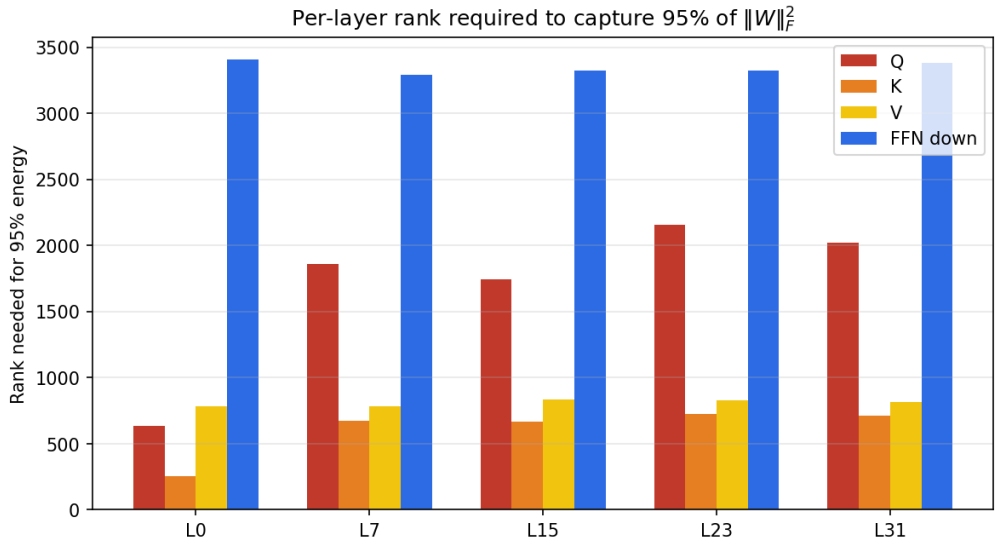

Figure 5: Per-layer rank required for 90%, 95% and 99% Frobenius-energy retention across the 32 transformer blocks of Llama-3.1-8B. The flat plateau between L=5 and L=28 shows that a *single* compression rank (here k=1024) is geometry-justified for the bulk of the network — the per-layer "what would be needed for 95%" band sits in the 1 400–1 800 range, and dropping to k=1024 trades a known small energy fraction for the cache-fit gain measured in Table 1. Edge layers $L \leq 4$ and $L \geq 29$ are spectrally stiffer and absorb most of the +13.30% PPL penalty.

Table 12: Decode tok/s by rank (n=8 samples per row, 4 prompts × 2 token lengths). Ratio is mean GRC throughput divided by the same-pack baseline mean.

Condition	Decode tok/s	n	Ratio vs. baseline
baseline	36.10 ± 0.09	8	1.0000
$k = 1024$	38.36 ± 0.18	8	1.0627
$k = 1536$	35.21 ± 3.35	8	0.9754
$k = 2048$	36.48 ± 1.21	8	1.0104

[25] F. Wilcoxon. "Individual Comparisons by Ranking Methods". In: *Biometrics Bulletin* 1.6 (1945), pp. 80–83.

[26] N. Elhage, T. Hume, C. Olsson, N. Schiefer, et al. *Toy Models of Superposition*. Anthropic Transformer Circuits Thread. 2022.

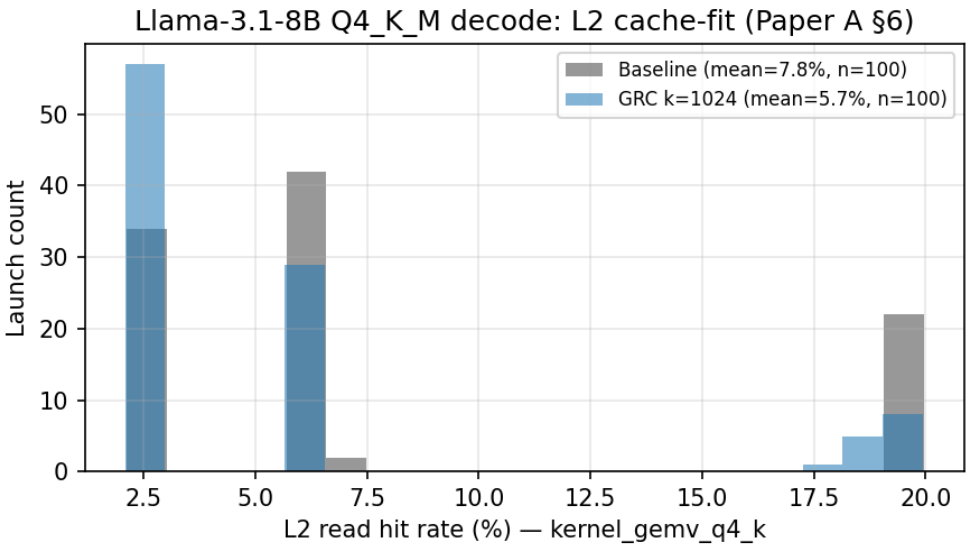

Figure 6: Direct Nsight Compute trace of L2 read hit-rate on the residual `kernel_gemv_q4_k` (FFN matmul, identical in both runs) for 100 representative decode-phase launches each on Llama-3.1-8B Q4_K_M, RTX 4070 Laptop (32 MB L2). Baseline mean 7.84%, GRC k=1024 mean 5.74% (Δ=−2.09 pp). The drop is consistent with GRC installing P_t (128 MB) and W_{proj} (72 MB) persist hints to keep the *compressed-attention* working set resident, evicting (uncompressed) FFN weights — the cost paid for the bandwidth reduction reported in Table 1. A like-for-like trace of the compressed-attention kernel itself is not possible because the kernel exists only when GRC is active.

Table 13: Decode tok/s grouped by generated-token budget (n=4 prompts per cell, mean±stdev). The k=1024 super-baseline is stable across both budgets; the k=1536 row inherits the per-prompt variance reported in Table 1.

Decode tok	Baseline	$k=1024$	$k=1536$	$k=2048$
128	36.17 ± 0.05	38.52 ± 0.05	34.15 ± 4.67	36.10 ± 1.40
256	36.02 ± 0.05	38.20 ± 0.00	36.27 ± 1.19	36.85 ± 1.04

Table 14: Validation gates on pack `whitepaper_pack_20260427_121815`.

Gate	**Threshold**	**Measured**	**Result**
k=1024 decode retention	≥ 95%	106.27%	PASS
k=1536 decode retention	≥ 75%	97.55%	PASS
k=1536† warmup-proxy decode	≥ 75%	101.04%	PASS
k=1536† warmup-proxy prefill	≤ 225%	108.48%	PASS
coding/256 lower-95 retention	≥ 67%	86.60%	PASS
reasoning/256 lower-95 retention	≥ 67%	85.64%	PASS
PPL delta (WikiText-2)	≤ +15%	+13.30%	PASS

Table 15: Honest accounting of GRC's benefits and limits, distilled from the v0.6.0 technical report. Severities reflect deployment impact.

Dimension	Sign	Severity
No calibration data needed (pure weight geometry)	+	Major
Super-baseline at k=1024 (106.27% decode)	+	Major
Near-lossless throughput at k=1536	+	Moderate
Deterministic, cacheable, sign-canonical basis	+	Moderate
Sub-second warm startup (small models)	+	Moderate
Zero training infrastructure	+	Major
Modular (Q/K/V only; FFN untouched)	+	Moderate
+13.30% WikiText-2 PPL at k=1536	–	Major
Prefill 8–15% slower (sequential path)	–	Moderate
`AXEX_MANIFOLD_K_MAX` hard cap at 1536	–	Moderate
Minimal VRAM reduction (+36 MiB delta)	–	Moderate
First-run cold start 60–120 s	–	Moderate
GRC throughput variance ~6× baseline	–	Moderate
No power savings (same 109 W TDP)	–	Mild
$\mathbf{W}_O$ excluded; FFN excluded	–	Mild
Single hardware, single model validated	–	Major
CUDA-only (no ROCm/Metal/CPU fallback)	–	Mild

Table 16: Acknowledged methodological gaps. Items 1, 3, 4 require infrastructure outside the scope of this independent project; item 2 is on the near-term roadmap.

Gap	Missing	Why it matters
Direct L2 cache-hit measurement	No Nsight Compute counter trace; cache-fit argued from access-pattern arithmetic.	Alternative micro-arch explanations (register pressure, scheduler) cannot be ruled out without counters.
Task-level evaluations	Only WikiText-2 PPL; no MMLU/GSM8K/HumanEval/IFEval.	+13.30% PPL is a structural signal, not a behavioural one.
Head-to-head vs. AWQ / GPTQ / SmoothQuant	No direct A/B on identical hardware; we sit on the same Q4_K_M baseline.	Production-scale ranking unverifiable without compatible-runtime baselines.
Cross-hardware validation	All measurements on RTX 4070 Laptop; Table 2 is calculated, not measured.	Cache-fit cannot be claimed as general — only as observed on this GPU.

Table 17: Project phase status as of v0.6.0.

Phase	Objective	Status
1	Eliminate measurement instability	**Complete** (root cause: thermal throttling; 30 s cooldown fix)
2	Validate under locked protocol	**Complete** (7/7 gates pass)
3	Cross-model transfer (Mistral-7B-v0.1)	In progress
3	Cross-hardware transfer	Blocked (single-GPU constraint)
4	External reproduction package	**Complete** (`repro/REPRODUCE.md`)